

Graph Theory Assignment 2 Report

Group 5

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Group Homework

Adapted from MIT Urban Operations Research, with modifications

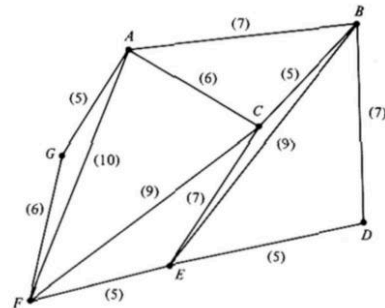
You are given this graph network on the right side

You must:

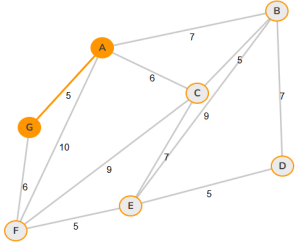
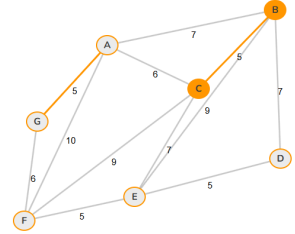
1. Implement Prim's algorithm to find MST
2. Implement Kruskal's algorithm to find MST
3. Independently find another algorithm than any explained on this slide, then implement it to find MST
4. Simulate if any node/edge failed (cannot be traversed), report how your algorithm adapt

You must submit

- A. GitHub repository containing the implementation
- B. A PDF report of 1-4, committed in the repository



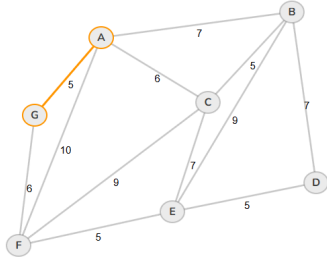
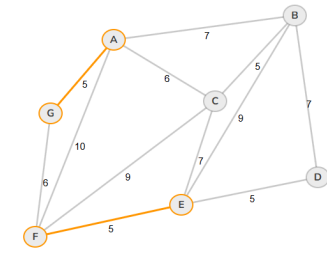
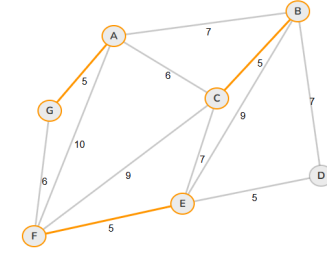
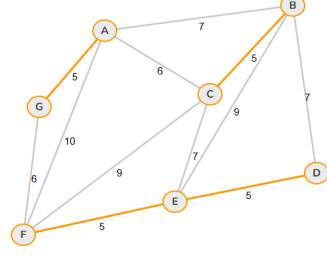
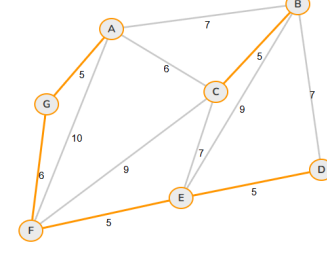
Boruvka

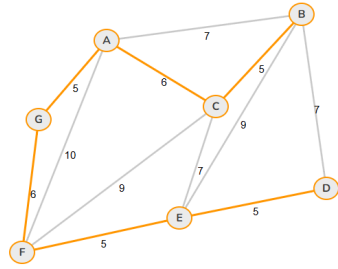
Step	U	Edg(u, v)	V - U	Graph
0	{ }	-	{ A ,B ,C ,D ,E ,F ,G }	
1	{A, G}	(A - G) = 5	{B, C, D, E, F}	
2	{A, G} {B, C}	(B - C) = 5	{D, E, F}	

3	{A, G}, {B, C}, {D, E}	(D - E) = 5	{F}	
4	{A, G}, {B, C}, {D, E, F}	(E - F) = 5	{}	
5	{A, B, C, G}, {D, E, F}	(A - C) = 6	{}	
6	{A, B, C, D, E, F, G}	(F - G) = 6	{}	

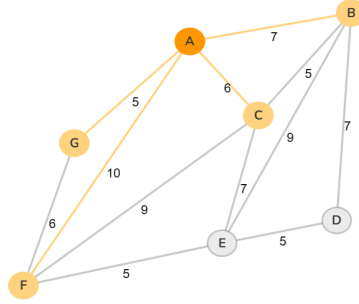
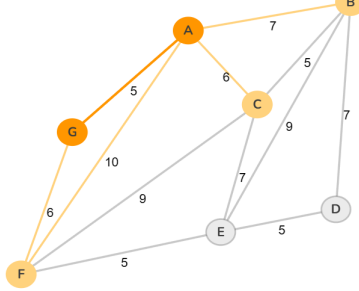
Kruskal

Step	U	Edg(u, v)	V - U	Graph
0	{}	-	{ A ,B ,C ,D ,E ,F ,G }	-

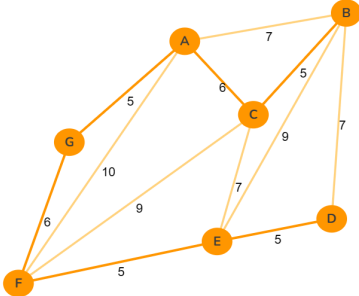
1	{ A , G }	$(A - G) = 5$	{ B , C , D , E , F }	
2	{ A , G } , { B , C }	$(B - C) = 5$	{ D , E , F }	
3	{ A , G } , { B , C } , { D , E }	$(D - E) = 5$	{ F }	
4	{ A , G } , { B , C } , { D , E , F }	$(E - F) = 5$	{ }	
5	{ A , B , C , G } , { D , E , F }	$(A - C) = 6$	{ }	

6	{ A , B , C , D , E , F , G }	(F - G) = 6	{ }	
7	{ A , B , C , D , E , F , G }	(A - B) = 7 cycle(B - D) = 7 cycle(C - E) = 7 cycle(B - E) = 9 cycle(C - F) = 9 cycle(A - F) = 10 cycle	{ }	

Prim

Step	U	Edg(u, v)	V - U	Graph
0	{ A }	-	{ B , C , D , E , F , G }	
1	{ A , G }	(A - G) = 5	{ B , C , D , E , F }	

2	{ A, G, C }	$(A - C) = 6$	{ B, D, E, F }	
3	{ A, G, C, B }	$(C - B) = 5$	{ D, E, F }	
4	{ A, G, C, B, F }	$(G - F) = 6$	{ D, E }	
5	{ A, G, C, B, F, E }	$(F - E) = 5$	{ D }	

6	{ A, G, C, B, F, E, D }	$(E - D) = 5$	{ }	
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